



**SIMATS**  
ENGINEERING



**SIMATS**  
Saveetha Institute of Medical And Technical Sciences  
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

**ITA0510 – COMPUTER VISION**

**EXPERIMENT NO: 18**

**Name: Premarubha N**

**Reg No: 192525020**

### **18. Perform Edge detection using Sobel Matrix along Y axis**

#### **AIM:**

Edge detection using Sobel Matrix along Y axis

#### **PROGRAM:**

```
import cv2

img = cv2.imread(r"C:\Users\premarubha\Desktop\cv model image.jpg")

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

sobely = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=3)

sobely = cv2.convertScaleAbs(sobely)

cv2.imshow("Original Image", img)

cv2.imshow("Sobel Y Edge Detection", sobely)

cv2.imwrite(r"C:\Users\premarubha\Desktop\sobel_y_image.jpg", sobely)

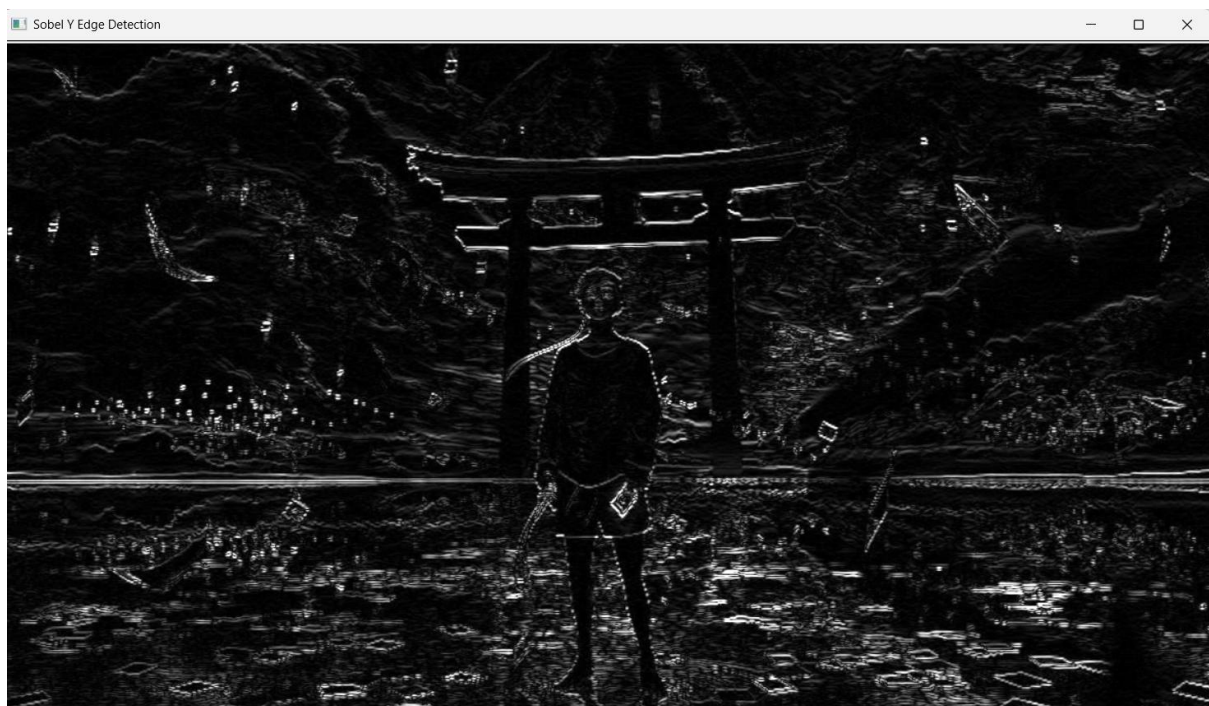
cv2.waitKey(0)

cv2.destroyAllWindows()
```

**INPUT:**



**OUTPUT:**



**RESULT:**

Thus the program is executed successfully.